CMPE 460 - Deep Learning

Assignment 01
Deadline: 18.10.2026 23.50

RULES

- This is an individual study. Each student is required to show the originality of their submission. Submissions involved in plagiarism will be penalized.
- Submit your implementations and trained models in a zip file named [**student_id_assignment01**]**.zip**.
- Late submissions will **not** be accepted. The submission system will be closed after the deadline. Submissions by email will **not** be accepted.

Submission

- Your submission must follow the directory tree given below

```
.
├── data
|   ├── train.csv
|   └── validation.csv
├── saved_objects
|   └── model.pt
├── main.py
├── run_test.py
├── model.py
├── trainer_utils.py
├── data_utils.py
└── README.md
```

If it is necessary, you can delete `data` folder while uploading your submission. Make sure that you provide the necessary information/ links in `README.md` file.

ASSIGNMENT

Your task is to train a **Fully-Connected Feedforward Neural Network**[1] for a seven-class classification problem, i.e., discriminating seven types of dry beans based on grain features such as form, shape, and structure.

Training and validation sets, along with evaluation codes, are shared in Learn. Details about the codes are shared in the files. Follow the instructions in the files; do not modify the evaluation codes. You can run the `run_test.py` script to evaluate your models. During the evaluation, similar code will be executed. If your code fails in this script, you may not get points for the related tasks.

1. ***(25 points)*** Implement a Dataset class (`data_utils.py`) to read the given dataset and apply the necessary preprocessing operations.

2. ***(25 points)*** Implement a Model class (`model.py`) utilizing a Fully-Connected Feedforward Neural Network architecture.

3. ***(25 points)*** Train your model with different hyperparameters and monitor the training progress. Share your experiments in the Weights and Biases Report[2]. Make sure that you provide the publicly available access link in your `README.md` file.

4. The remaining points for this assignment will be granted according to the performance of the models in the test set[3]. Each model will be ranked relative to the submissions in the class. For example, if model's performance is ranked in the 3rd quartile, you will get 5 points additional to the points given above.

F1-Score in Quartile	**Points**
1st Quartile	25 points
2nd Quartile	15 points
3rd Quartile	5 points
4th Quartile	0 point

[1] In this assignment, only the Feedforward Neural Network models are allowed. Submissions involving different model architectures will not be accepted.

[2] https://docs.coreweave.com/products/wandb/reports

[3] Baseline F1 score: 0.808334